

Mapping Metabolic Individuality: Genome-Scale Models Capture Stable, High-Dimensional Liver Flux Signatures Beyond Diet, Sex, and Age

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1 Supplementary Methods

1.1 Transcriptomic preprocessing and TPM conversion

Prior to integration with the GEM, transcriptomic expression values, originally provided as $\log_2(\text{RPKM} + 1)$, were converted to TPM to obtain expression measures suitable for context-specific model reconstruction.¹ First, log-transformed expression values were back-transformed to RPKM values using

$$\text{RPKM} = 2^x - 1, \quad (1)$$

where x denotes the reported $\log_2(\text{RPKM} + 1)$ value. TPM values were then calculated by column-wise normalization,

$$\text{TPM}_{g,s} = \frac{\text{RPKM}_{g,s}}{\sum_{g'} \text{RPKM}_{g',s}} \times 10^6, \quad (2)$$

for each sample s independently.

1.2 Consolidation of duplicate omics identifiers

Following the transformation, duplicate gene identifiers in the transcriptomic dataset were combined by averaging numeric expression values across matching entries, yielding a single expres-

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sion value per gene consistent with the gene-reaction rules defined in mouseGEM. An identical procedure was applied to the proteomic dataset to consolidate duplicate protein identifiers. The resulting collapsed omics datasets were used for context-specific GEM reconstruction. By contrast, the omics-based classification analyses described later used the corresponding raw datasets, that is, without collapsing duplicate identifiers.

1.3 Transcriptomics gene name harmonization

To enable integration of the transcriptomic dataset with mouseGEM, a gene-identifier harmonization pipeline was applied to the transcriptomic table. In both datasets, identifiers were given as mouse gene symbols, but nomenclature differences remained between the omics data and mouseGEM. An initial set of directly matched genes was obtained by intersecting the gene identifiers in mouseGEM and the transcriptomic dataset. Genes present in mouseGEM but absent from the transcriptomic dataset after this matching step were then taken for additional processing.

To resolve nomenclature discrepancies, the Python `mygene` library² was used to query the MyGene.info web service. The unmatched mouseGEM gene list was queried against the `symbol` and `alias` scopes for *Mus musculus*. When multiple records were returned for a query, all returned aliases were combined into a single synonym set for that mouseGEM gene. Transcriptomic identifiers were then compared against these synonym sets. If a transcriptomic identifier matched a synonym associated with a mouseGEM gene, the identifier in the omics table was replaced with the corresponding mouseGEM gene name.

A proposed mapping was rejected if it introduced a nomenclature collision. Here, a collision was defined as a case in which a transcriptomic identifier was mapped to a mouseGEM gene name that had already been assigned to a different transcriptomic identifier during harmonization. In such cases, the mapping was not applied and the original transcriptomic identifier was retained.

Following automated mapping, 28 still-unmatched mouseGEM genes were manually curated to identify potential matches in the transcriptomic gene list using external databases, primarily NCBI Gene.³ Curated mappings were applied only when they did not introduce collisions as defined above. Identifiers that remained unmatched after automated and manual harmonization were left unchanged in the omics table. The harmonized transcriptomic table was then

used as input for downstream ftINIT reconstruction and E-flux-based constraint generation. MyGene.info was accessed on 21 October 2025.

1.4 Proteomics gene name harmonization

Proteomic identifiers were provided as mouse gene symbols. To enable integration of the proteomic dataset with mouseGEM, a harmonization pipeline analogous to that used for transcriptomics was applied to the proteomics table. Because the number of unmatched identifiers was higher than in the transcriptomic dataset, no manual curation step was performed.

Primary querying was carried out using the `mygene` library as described above. For mouseGEM genes that remained unmatched after this step, additional synonyms were retrieved from NCBI Gene using the `Bio.Entrez` module from Biopython.⁴ Each identifier was searched with organism restriction to *Mus musculus*, the matching record was retrieved in XML format, and synonyms were extracted from the `Gene-ref_syn` field. Proteomic identifiers were then compared against the resulting synonym sets, and valid non-conflicting matches were relabeled with the corresponding mouseGEM gene names in the proteomics table.

As above, mappings were rejected if they introduced a nomenclature collision, that is, if a mouseGEM gene name had already been assigned to a different proteomic identifier during harmonization. Identifiers that remained unmatched after automated harmonization were left unchanged in the proteomics table. The harmonized proteomics table was then used for downstream ftINIT reconstruction. MyGene.info and NCBI Gene were accessed on 21 October 2025.

1.5 Preparation of mouseGEM for ftINIT

As an initial step for ftINIT reconstruction, mouseGEM was prepared using the `prepINITModel` function in the RAVEN Toolbox,⁵ with ftINIT executed in MATLAB through the same framework. Preparation required the reference model together with lists of essential metabolic tasks and spontaneous reactions provided with mouseGEM. The task list, derived from the humanGEM framework,⁶ consisted of generic metabolic functions expected to be feasible in all cells. Spontaneous reactions were included as specified in the mouseGEM resources. All remaining `prepINITModel`

parameters were left at default values. The resulting task-feasible minimal model was subsequently reduced using `simplifyModel`⁵ with default settings prior to ftINIT reconstruction.

1.6 Formatting transcriptomic and proteomic evidence for ftINIT

TPM-transformed transcriptomic data were formatted into an ftINIT-compatible object with `genes` containing gene identifiers, `tissues` containing mouse identifiers, `celltypes` left empty, and `levels` containing the TPM matrix. For reaction scoring, the expression threshold was set to the global median TPM value (5.4). A global threshold was preferred over mouse-specific thresholds because all samples originated from the same tissue type, and per-sample thresholding could have introduced artificial between-model differences. The median was chosen as a relatively conservative cutoff to avoid overly permissive reaction inclusion.

Proteomics data were formatted analogously. Quantitative protein abundances were discretized using the 25th, 50th, and 75th percentiles of the full dataset. Values above the 75th percentile were labeled *Strong*, values between the 50th and 75th percentiles *Moderate*, values between the 25th and 50th percentiles *Weak*, and values below the 25th percentile *Negative*. Assignments were stored as a sparse matrix for ftINIT compatibility.

During reconstruction, ftINIT assigns evidence scores to reactions for each mouse liver sample. Transcriptomic evidence is incorporated as a continuous expression-based score relative to the selected threshold, whereas proteomic evidence is incorporated as categorical abundance-based evidence. Reaction scores are integrated through gene-protein-reaction rules, while reactions lacking gene associations are handled using the default ftINIT scoring scheme. Full implementation details are described in.⁷

1.7 Context-specific reconstruction with ftINIT

The processed mouseGEM together with formatted transcriptomic and proteomic evidence was used as input to ftINIT. Reconstruction employed the '1+1' configuration, which applies stronger pruning, including removal of more reactions lacking GPR rules, than the '1+0' setting and therefore typically yields smaller context-specific GEMs. The `celltype` argument was left empty and no metabolomics constraints were supplied. All remaining parameters were kept at default values, including gene pruning (`removeGenes = true`) and score-weighted task fitting (`useScoresForTasks = true`), which uses reaction evidence scores during task-enforcement

106 gap filling.

107 **1.8 Application of transcriptomics-derived E-flux constraints**

108 Following ftINIT reconstruction, mouse liver GEMs were further constrained using an E-flux-
109 style approach⁸ to incorporate quantitative transcriptomic information into reaction bounds.
110 Whereas ftINIT determines reaction inclusion, E-flux modulates the allowable flux range of re-
111 tained reactions according to mouse-specific transcript abundance.

112 Transcript abundance was treated as a relative proxy for reaction capacity rather than a direct
113 estimate of enzyme-limited flux. For single-gene reactions, capacity was set equal to the TPM
114 value of the corresponding gene. For multi-gene GPR rules, capacities were derived by evalu-
115 ating the Boolean rule structure with parentheses preserved, interpreting AND relationships as
116 the minimum and OR relationships as the maximum of participating gene values. Genes absent
117 from the dataset or with non-positive expression were assigned zero expression.

118 If the resulting reaction capacity was zero or undefined, a floor value of 10^{-5} was imposed to
119 avoid numerical instability. Reactions lacking GPR rules were left unconstrained.

120 For reversible reactions, bounds were set symmetrically with lower and upper bounds equal to
121 the negative and positive TPM-derived capacities, respectively. For irreversible reactions, the
122 lower bound was set to zero and the upper bound to the TPM-derived capacity.

123 **1.9 Application of diet exchange constraints**

124 Dietary and physiological exchange constraints were applied identically across all mouse liver
125 GEMs. Dietary uptake upper bounds were derived from a standardized chow intake of 3 g/day
126 using nutrient mass fractions from the Teklad 2018 Global 18% Protein Rodent Diet datasheet.⁹
127 For each nutrient-associated exchange reaction r , daily intake was calculated as

$$g_{\text{day}}(r) = 3 \text{ g/day} \times \text{massFrac}(r), \quad (3)$$

128 and converted to an uptake upper bound in mmol/day as

$$\text{UB}_{\text{mmol/day}}(r) = \frac{g_{\text{day}}(r)}{\text{MW}(r)} \times 1000, \quad (4)$$

where $MW(r)$ denotes molecular weight in g/mol. This standardized scheme was intended to represent long-term nutrient availability rather than short-term feeding variation or mouse-specific intake. The overall strategy of combining diet-derived exchange bounds with expression-informed intracellular constraints follows previous liver GEM studies.¹⁰

Additional physiological exchange upper bounds were adopted largely from,¹⁰ with inclusion of choline,¹¹ lactate,¹² glycerol,¹³ sulfate,¹⁴ and phosphate (P_i).¹⁵ Exact values are provided in the supplementary data table.

The oxygen uptake constraint was fixed to a mmol/day value derived from a mass-specific oxygen consumption rate ($VO_2 = 1561 \text{ mL kg}^{-1} \text{ h}^{-1}$,¹⁰) and a reference body weight of 25 g. Conversion to mmol/day assumed a molar gas volume of $22.414 \text{ L mol}^{-1}$.

Water uptake was left unconstrained, whereas all remaining uptake reactions were fixed to zero.

1.10 Detailed visualization procedures

For flux-feature visualization, the flux matrix $\mathbf{X}$ was standardized using `StandardScaler`¹⁶ prior to dimensionality reduction. Two-dimensional t-SNE was performed in Python with `sklearn.manifold.TSNE`¹⁶ using perplexity 30, random initialization, and 1,000 iterations, while all other parameters remained at default values. UMAP was applied to the same standardized matrix using `umap-learn`¹⁷ with 25 neighbors, minimum distance 0.4, Manhattan distance, and a fixed random state. Embeddings were visualized as scatter plots colored by mouse identity or diet group.

Diet-based visualization of the the flux matrix $\mathbf{X}$ used the same t-SNE and UMAP settings described above, with embeddings colored by diet group.

1.11 Random forest classification procedures

Random forest (RF) classification was performed in Python using `scikit-learn's RandomForestClassifier`.^{16,18} RF was selected because it captures nonlinear decision boundaries and feature interactions while providing internal feature-importance estimates. Unless otherwise stated, models were trained using a `scikit-learn Pipeline`,¹⁶ randomized hyperparameter search with `RandomizedSearchCV`,^{16,19} stratified 80/20 train-test splitting, and repeated resampling. Hyperparameters were optimized by inner cross-validation on the training set, after which the best model was refit on the full train-

ing data and evaluated on the held-out test set. Performance was summarized across repetitions using averaged confusion matrices and macro-averaged metrics, including macro- F_1 , accuracy, and macro-recall; macro-precision was additionally reported for mouse-identity prediction from the flux matrix $\mathbf{X}$.

For mouse-identity prediction from the flux matrix $\mathbf{X}$, reaction-level flux values were used as features and mouse identifiers as labels. Hyperparameters were optimized using nested resampling with an outer stratified 80/20 train-test split and an inner 4-fold stratified cross-validation loop, ranking models by mean macro- F_1 . Five hyperparameter configurations were sampled per repetition from a search space including number of trees (50–100), maximum depth (10, 20, 30), minimum samples per split (2–10), and minimum samples per leaf (1–5). Because this procedure was computationally expensive, an additional analysis used fixed hyperparameters with repeated stratified 80/20 resampling. In this setting, the classifier used 200 trees, maximum depth 18, minimum samples per split 8, minimum samples per leaf 2, and square-root feature subsampling. The model seed was fixed, while variability across runs was introduced through repeated train-test resampling. This procedure was repeated 50 times, and confusion matrices were averaged across runs.

Diet group (CD vs HF) was predicted from the PA matrix using the same workflow. Owing to the lower-dimensional feature space, the hyperparameter search range was expanded to 100–500 trees, maximum depth values of none, 10, 20, 30, 40, or 50, minimum samples per split of 2–10, and minimum samples per leaf of 1–5, with 30 hyperparameter configurations sampled per run. The full procedure was repeated 50 times.

Diet prediction from the flux matrix $\mathbf{X}$ used the same search space, optimization strategy, and evaluation metrics as the mouse-identity analysis, but incorporated group-aware splitting to prevent leakage from multiple samples belonging to the same mouse. Mouse identity was therefore used as a grouping variable so that all samples from a given mouse were assigned exclusively to either the training or test partition. Outer train-test splits were generated with `GroupShuffleSplit`,¹⁶ and inner hyperparameter tuning used 4-fold `StratifiedGroupKFold`.¹⁶ Randomness across repetitions was introduced by varying the outer split seed, whereas the RF estimator used a fixed random seed. The full nested procedure was repeated 30 times. To identify difficult individuals, the proportion of misclassified samples was calculated for each mouse across held-out test evaluations.

Diet group was also predicted from the transcriptomic feature matrix using the same workflow as for the PA matrix, except that the randomized search budget was reduced to 10 and the procedure repeated 30 times. For these analyses, raw transcriptomic data were used prior to consolidation of duplicate identifiers.

2 Software

Computational analyses were performed using MATLAB R2025b (version 25.2.0.2998904) and Python 3.12.10.

In MATLAB, the COBRA Toolbox (version 2.51.0) was used for model handling, FBA, blocked reaction detection, and flux sampling,²⁰ and the RAVEN Toolbox (version 2.10.5) was used for prepINITModel, ftINIT, and related reconstruction utilities.⁵ The Parallel Computing Toolbox (version 25.2, R2025b) was used for parallelised MATLAB computations. Gurobi Optimizer (version 12.0.3) was used as the LP/MILP solver.²¹ See Supplementary Material

Default Gurobi tolerance settings were used: FeasibilityTol = 10^{-6} , OptimalityTol = 10^{-6} , IntFeasTol = 10^{-5} , MIPGap = 10^{-4} , and MIPGapAbs = 10^{-10} .

In Python, preprocessing, identifier harmonisation, statistical analysis, machine learning, and visualisation were performed using mygene,² biopython,⁴ scikit-learn,¹⁶ umap-learn,¹⁷ scikit-bio,²² SciPy,²³ NumPy,²⁴ pandas,²⁵ matplotlib,²⁶ and seaborn.²⁷

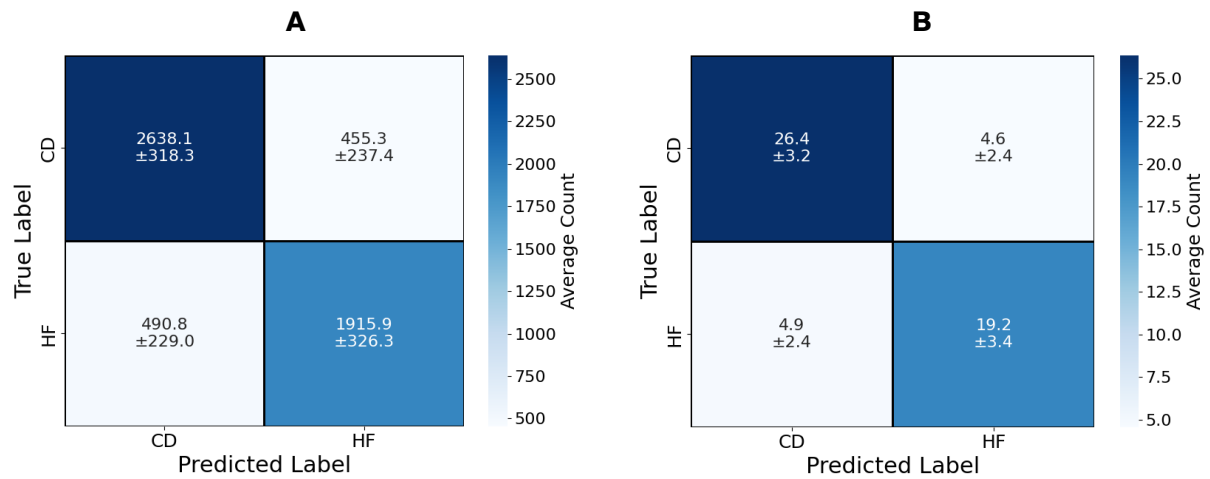


Figure S1: Confusion matrices for diet classification using RF models. (A) Predictions at the individual sample level. (B) Predictions at the mouse level obtained by majority voting across samples from each mouse. Rows represent true diet labels and columns predicted labels. Values show mean counts across repeated runs with standard deviations below each mean.

3 Supplementary results

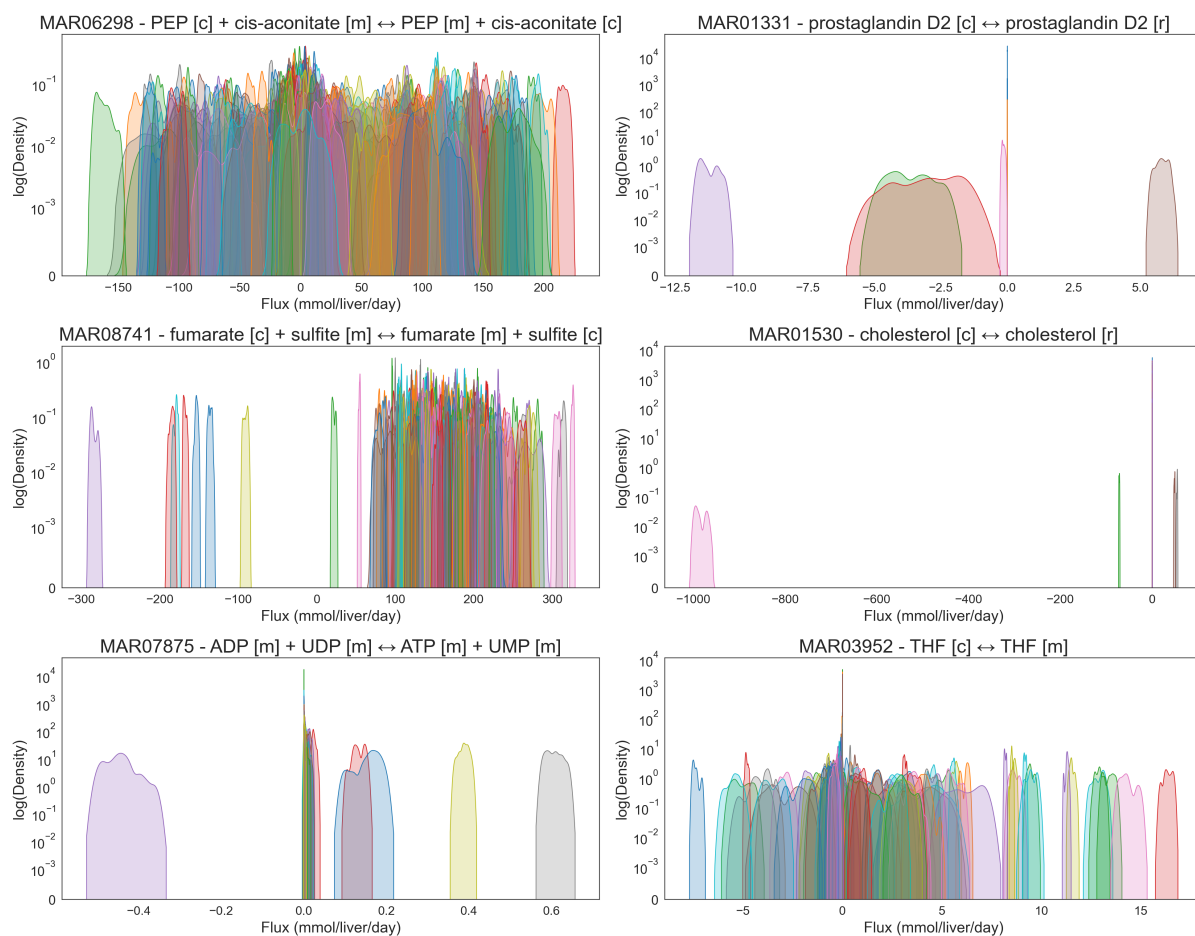


Figure S2: Complete set of per-mouse flux density distributions for the selected high-importance reactions identified by the RF classifier of mouse identity from the flux matrix $\mathbf{X}$. Each panel corresponds to one reaction, with curves colored by mouse.

Table S1: Reaction-level summary for the glycolysis / gluconeogenesis pathway. Reaction identifiers are mouseGEM reaction IDs. The table reports the fraction of mice in which each reaction was present and its mean random forest feature importance.

Rank	mouseGEM ID	Reaction name	Fraction present	Avg. importance
1	MAR04099	Acetate:CoA ligase (AMP-forming)	1.000000	0.00000005020018
2	MAR04281	(S)-Lactate:NAD ⁺ oxidoreductase	1.000000	0.0003673128
3	MAR04388	(S)-Lactate:NAD ⁺ oxidoreductase	1.000000	0.00005893098
4	MAR08357	Acetaldehyde:NAD ⁺ oxidoreductase	1.000000	0.00007789027
5	MAR04379	ATP:D-fructose-6-phosphate 1-phosphotransferase	1.000000	0.0000896069
6	MAR04301	UTP:D-fructose-6-phosphate 1-phosphotransferase	1.000000	0.0003779427
7	MAR04355	sedoheptulose 1,7-bisphosphate D-glyceraldehyde-3-phosphate-lyase	1.000000	0.00000002209691
8	MAR04358	ATP:pyruvate 2-O-phosphotransferase	1.000000	0.0005334648
9	MAR04363	2-phospho-D-glycerate hydro-lyase (phosphoenolpyruvate-forming)	1.000000	0.00114748
10	MAR04365	D-phosphoglycerate 2,3-phosphomutase	1.000000	0.000972467
11	MAR04368	ATP:3-phospho-D-glycerate 1-phosphotransferase	1.000000	0.0009298757
12	MAR04370	3-Phospho-D-glyceroyl phosphate phosphohydrolase	1.000000	0.00000001920032
13	MAR04371	3-phospho-D-glycerate 1,2-phosphomutase	1.000000	0
14	MAR04372	2,3-bisphospho-D-glycerate 2-phosphohydrolase	1.000000	0.00000005765784
15	MAR04373	D-glyceraldehyde-3-phosphate:NAD ⁺ oxidoreductase (phosphorylating)	1.000000	0.001022629
16	MAR04375	D-fructose-1,6-bisphosphate D-glyceraldehyde-3-phosphate-lyase (glycerone-phosphate-forming)	1.000000	0.0007177482
17	MAR04377	D-Fructose-1,6-bisphosphate 1-phosphohydrolase	1.000000	0.0000001663364
18	MAR04381	D-glucose-6-phosphate aldose-ketose-isomerase	1.000000	0.0003557251
19	MAR04394	ATP:D-glucose 6-phosphotransferase	1.000000	0.00005004299
20	MAR04396	alpha-D-glucose 1,6-phosphomutase	1.000000	0.0005343452
21	MAR04521	D-Glucose-6-phosphate phosphohydrolase	1.000000	0.00000006082996
22	MAR06412	acetyl-CoA:enzyme N6-(dihydrolipoyl)lysine S-acetyltransferase	1.000000	0.0005657623
23	MAR01169	Glycerol-3-Phosphate Dehydrogenase (FAD), Mitochondrial	1.000000	0
24	MAR20069	MAR20069	1.000000	0.000556628
25	MAR04097	Acetate:CoA ligase (AMP-forming)	0.901818	0.00007671558
26	MAR04108	acetyl adenylate:CoA acetyltransferase	0.901818	0.00001948662
27	MAR04133	ATP:acetate adenyltransferase	0.901818	0.00001192951
28	MAR07747	ADP:D-glucose 6-phosphotransferase	0.647273	0.00009275564
29	MAR04283	Acetaldehyde:NADP ⁺ oxidoreductase	0.134545	0.0003997301
30	MAR04391	D-glyceraldehyde-3-phosphate aldose-ketose-isomerase	0.003636	0.000402353

Table S2: Reaction-level summary for the tyrosine metabolism pathway. Reaction identifiers are mouseGEM reaction IDs. The table reports the fraction of mice in which each reaction was present and its mean random forest feature importance.

Rank	mouseGEM ID	Reaction name	Frac. present	Avg. imp.
1	MAR04681	L-phenylalanine carboxy-lyase (phenylethylamine-forming)	1.000000	0.000066
2	MAR04682	Phenethylamine:oxygen oxidoreductase (deaminating)	1.000000	0.000066
3	MAR03935	S-(hydroxymethyl)glutathione formaldehyde-lyase (glutathione-forming)	0.992727	0.000987
4	MAR08530		0.989091	0.001456
5	MAR12020	Dt-Diaphorase: NADP Quinone Oxireductase	0.930909	0.000559
6	MAR12021	Dt-Diaphorase: NAD Quinone Oxireductase	0.930909	0.000570
7	MAR12019	Leukoaminochrome Autoxidation	0.494545	0.000774
8	MAR12022	2, 3-Dihydro-1H-Indole-5, 6-Dione Tautomerization	0.476364	0.000967
9	MAR04683	Phenylacetaldehyde:NAD ⁺ oxidoreductase	0.389091	0.000498
10	MAR04685	Phenylacetaldehyde:NAD ⁺ oxidoreductase	0.345455	0.000414
11	MAR04687	Aldehyde:NADP ⁺ oxidoreductase	0.305455	0.000439
12	MAR04686	Aldehyde:NADP ⁺ oxidoreductase	0.280000	0.000307
13	MAR04702		0.021818	0.000190
14	MAR04703	Gentisate aldehyde:NAD ⁺ oxidoreductase	0.021818	0.000127
15	MAR04704		0.021818	0.000019
16	MAR08534		0.018182	0.000025
17	MAR08535		0.018182	0.000047
18	MAR08537	L-tyrosine,tetrahydrobiopterin:oxygen oxidoreductase(3-hydroxylating)	0.003636	0.000252
19	MAR08540		0.003636	0.000255
20	MAR03599		0.003636	0.000271
21	MAR11467	Formation of Vanilpyruvic Acid from Vanilalanine	0.003636	0.000186
22	MAR11468	Formation of Vanil-Lactate	0.003636	0.000239
23	MAR12024	Tyrosinase: 5, 6-Dihydroxyindole Oxygen Oxidoreductase	0.003636	0.000220

Table S3: Transcriptomics RF classification of diet performance across repeated runs.

Metric	Mean	SD	Minimum	Maximum
Accuracy	0.919394	0.038104	0.854545	1.000000
Macro Precision	0.925050	0.034759	0.853333	1.000000
Macro Recall	0.915889	0.040511	0.840000	1.000000
Macro F1	0.917781	0.039395	0.845938	1.000000
Weighted Precision	0.923366	0.035530	0.854545	1.000000
Weighted Recall	0.919394	0.038104	0.854545	1.000000
Weighted F1	0.918769	0.038762	0.849249	1.000000
Balanced Accuracy	0.915889	0.040511	0.840000	1.000000
Cohen's Kappa	0.836213	0.078036	0.698630	1.000000
Specificity	0.954444	0.037634	0.866667	1.000000
Sensitivity	0.877333	0.077144	0.680000	1.000000

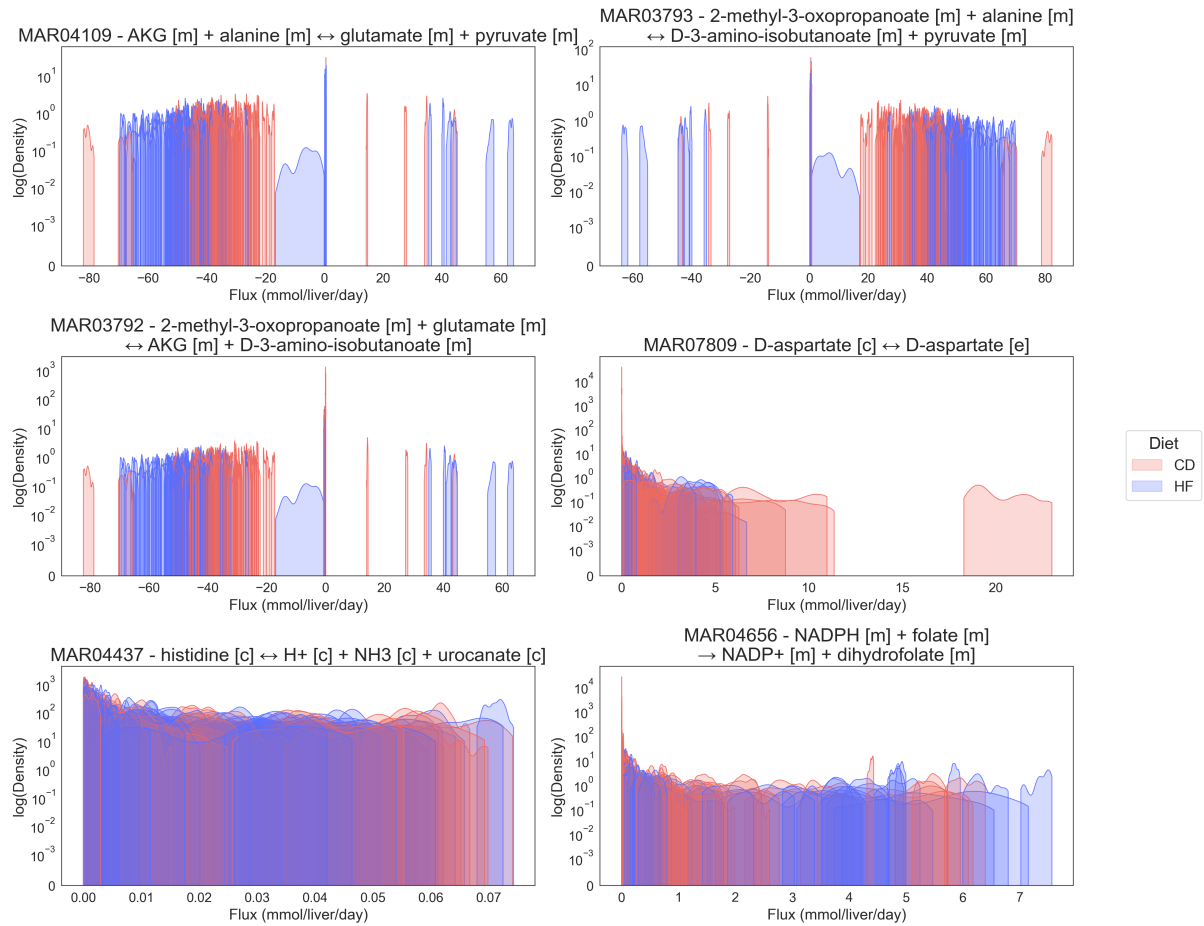


Figure S3: Complete set of reaction-level flux density distributions for the top 6 reactions predictive of diet from the flux matrix the flux matrix $\mathbf{X}$. Each panel shows one reaction, with density curves colored by diet group (CD in red; HF in blue).

Table S4: Reaction-level results for the peroxisomal beta oxidation of unsaturated fatty acids (n-7) pathway. Reaction identifiers are mouseGEM reaction IDs. Negative Spearman ρ values indicate relatively higher sampled flux in chow diet (CD), although none of these associations reached significance after FDR correction.

Rank	mouseGEM ID	Avg. importance	Spearman ρ	FDR
1	MAR03363	0.001034	-0.151376	0.116315
2	MAR03358	0.000855	-0.151376	0.116315
3	MAR03362	0.000754	-0.151376	0.116315
4	MAR03357	0.000503	-0.151376	0.116315
5	MAR03359	0.000387	-0.151376	0.116315
6	MAR03356	0.000363	-0.151376	0.116315
7	MAR03361	0.000275	-0.151376	0.116315
8	MAR03360	0.000250	-0.151376	0.116315

Table S5: Reaction-level results for the bile acid recycling pathway. Reaction identifiers are mouseGEM reaction IDs. Positive Spearman ρ values indicate relatively higher sampled flux in high-fat diet (HF), whereas negative values indicate relatively higher sampled flux in chow diet (CD).

Rank	mouseGEM ID	Avg. importance	Spearman ρ	FDR
1	MAR01888	0.001013979	0.000368	0.997357
2	MAR01893	0.000822727	-0.079538	0.509067
3	MAR01866	0.000808806	0.260533	0.000786
4	MAR01891	0.000695414	0.086811	0.470650
5	MAR01894	0.000606721	-0.032404	0.807984
6	MAR01864	0.000494347	0.278645	0.000261
7	MAR01886	0.000487646	0.074935	0.526334
8	MAR01890	0.000444246	0.045200	0.722439
9	MAR01867	0.000419847	0.279733	0.000243
10	MAR01863	0.000359038	0.276710	0.000282
11	MAR01892	0.000352544	-0.025776	0.867779
12	MAR01865	0.000325888	0.276589	0.000282
13	MAR01877	0.000314264	-0.021542	0.884725
14	MAR01889	0.000312773	-0.106419	0.339451
15	MAR01862	0.000306215	0.266916	0.000524
16	MAR01883	0.000268678	0.140665	0.157354
17	MAR01882	0.000256664	-0.121701	0.246735
18	MAR01879	0.000236579	0.130538	0.201018
19	MAR01881	0.000194782	0.021357	0.885393
20	MAR01885	0.000192914	-0.126396	0.224278
21	MAR01884	0.000178428	0.048054	0.701130
22	MAR01887	0.000152560	0.004143	0.978259
23	MAR01876	0.000145221	-0.022186	0.881180
24	MAR01880	0.000126013	0.127316	0.218448
25	MAR01875	0.000086179	0.018596	0.909033
26	MAR01874	0.000076173	0.242721	0.002160
27	MAR01847	0.000054614	-0.005892	0.966579
28	MAR01872	0.000001458	0.078433	0.511718
29	MAR01896	0.000000626	0.025868	0.867779
30	MAR01870	0.000000399	0.080459	0.501945
31	MAR01848	0.000000268	-0.008838	0.949186
32	MAR01868	0.000000260	0.052841	0.672449
33	MAR01853	0.000000148	0.045845	0.721858

Table S6: Reaction-level results for the androgen metabolism pathway. Reaction identifiers are mouseGEM reaction IDs. Negative Spearman ρ values indicate relatively higher sampled flux in CD, whereas positive values indicate relatively higher sampled flux in HF.

Rank	mouseGEM ID	Avg. importance	Spearman ρ	FDR
1	MAR01967	0.001767	-0.238429	0.002595
2	MAR01968	0.001559	-0.266133	0.000531
3	MAR01971	0.000924	-0.266133	0.000531
4	MAR01970	0.000874	-0.238429	0.002595
5	MAR01945	0.000592	-0.217721	0.007547
6	MAR01983	0.000347	-0.217746	0.007547
7	MAR02015	0.000324	-0.117872	0.269209
8	MAR02017	0.000284	-0.138478	0.164457
9	MAR01959	0.000271	-0.057546	0.651712
10	MAR01960	0.000270	-0.070803	0.553406
11	MAR01982	0.000265	-0.217746	0.007547
12	MAR01973	0.000251	-0.104767	0.350051
13	MAR01958	0.000250	-0.201100	0.018242
14	MAR02019	0.000205	-0.150908	0.117799
15	MAR02020	0.000175	0.089478	0.451053
16	MAR01952	0.000146	-0.139284	0.163939
17	MAR01974	0.000131	-0.096693	0.398256
18	MAR01978	0.000123	-0.089614	0.450779
19	MAR01977	0.000122	-0.087592	0.465672
20	MAR01976	0.000103	-0.103342	0.359573
21	MAR01969	0.000099	-0.032117	0.811323
22	MAR02018	0.000095	-0.084492	0.481076
23	MAR02016	0.000092	-0.105046	0.347879
24	MAR01944	0.000082	-0.029551	0.835337
25	MAR02014	0.000072	0.113865	0.291429
26	MAR02025	0.000050	-0.089572	0.450779
27	MAR01963	0.000020	0.047502	0.706978
28	MAR01962	0.000016	0.052657	0.672449
29	MAR01953	0.000013	0.012428	0.933305

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